

Differential Equations Lesson 02

Question 1:

Solve the differential equation $y' = \frac{dy}{dx} = 14x - 5$

a) $\int \left(\frac{dy}{dx} \right) dx = \underline{\hspace{2cm}} ?$

b) $y = \underline{\hspace{2cm}} ?$

Question 2:

Solve the differential equation $y' = \frac{dy}{dx} = \frac{2}{3}x - 1$

a) $\int \left(\frac{dy}{dx} \right) dx = \underline{\hspace{2cm}} ?$

b) $y = \underline{\hspace{2cm}} ?$

Question 3:

Solve the differential equation $y' = \frac{dy}{dx} = 3y - 12$

a) $\int \left(\frac{1}{3y-12} \cdot \frac{dy}{dx} \right) dx = \underline{\hspace{2cm}} ?$

b) $y = \underline{\hspace{2cm}} ?$

Question 4:

Solve the differential equation $y' = \frac{dy}{dx} = \frac{3}{4}y - 9$

a) $\int \left(\frac{1}{\frac{3}{4}y - 9} \cdot \frac{dy}{dx} \right) dx = \underline{\hspace{2cm}} ?$

b) $y = \underline{\hspace{2cm}} ?$

Question 5:

Solve the differential equation $y' = \frac{dy}{dx} = \frac{3x^3}{2y}$

a) $2y \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $\frac{dy}{dx} = \frac{3x^2}{2y}$ is $\underline{\hspace{2cm}} ?$

Question 6:

Solve the differential equation $y' = \frac{dy}{dx} = \frac{x^{1/2}}{5y}$

a) $5y \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $\frac{dy}{dx} = \frac{x^{1/2}}{5y}$ is $\underline{\hspace{2cm}} ?$

Question 7:

Solve the differential equation $y' = \frac{dy}{dx} = \sqrt[3]{x} \cdot y^2$

a) $\frac{1}{y^2} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $\frac{dy}{dx} = \sqrt[3]{x} \cdot y^2$ is $\underline{\hspace{2cm}} ?$

Question 8:

Solve the differential equation $y' = \frac{dy}{dx} = 4x^{-2/3} \cdot y^3$

a) $\frac{1}{y^3} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $\frac{dy}{dx} = 4x^{-2/3} \cdot y^3$ is $\underline{\hspace{2cm}} ?$

Question 9:

Solve the differential equation $y' = \frac{dy}{dx} = x^3 \cdot (5 + 4y)$

a) $\frac{1}{5 + 4y} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $\frac{dy}{dx} = x^3 (5 + 4y)$ is $\underline{\hspace{2cm}} ?$

Question 10:

Solve the differential equation $y' = \frac{dy}{dx} = x^{1/3} \cdot (5y - 4)$

a) $\frac{1}{5y - 4} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $\frac{dy}{dx} = x^{1/3} (5y - 4)$ is $\underline{\hspace{2cm}} ?$

Question 11:

Solve the differential equation $(14 + x^2)y' - 12xy = 0$

a) $\frac{1}{y} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $(14 + x^2)\frac{dy}{dx} - 12xy = 0$ is $\underline{\hspace{2cm}} ?$

Question 12:

Solve the differential equation $(4 + x^3)y' - 5x^2y = 0$

a) $\frac{1}{y} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) The general solution of the differential equation $(4 + x^3)\frac{dy}{dx} - 5x^2y = 0$ is $\underline{\hspace{2cm}} ?$

Question 13:

Solve the differential equation $y' \cdot \csc x - x \cdot \cot y = 0$

a) $\frac{1}{\cot y} \cdot \frac{dy}{dx} = \underline{\hspace{1cm} ? \hspace{1cm}}$

b) The general solution of the differential equation $y' \cdot \csc x - x \cdot \cot y$ is $\underline{\hspace{1cm} ? \hspace{1cm}}$

Question 14:

Solve the differential equation $x(x-3) \cdot y' - \sqrt{4-y^2} = 0$

a) $\frac{1}{\sqrt{4-y^2}} \cdot \frac{dy}{dx} = \underline{\hspace{2cm}}?$

b) The general solution of the differential equation $x(x-3) \cdot y' - \sqrt{4-y^2} = 0$ is $\underline{\hspace{2cm}}?$

Integration Formula:

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1} \left(\frac{u}{a} \right) + C$$

Hint: $\frac{1}{x(x+3)} = \frac{1/3}{x} - \frac{1/3}{x+3}$

